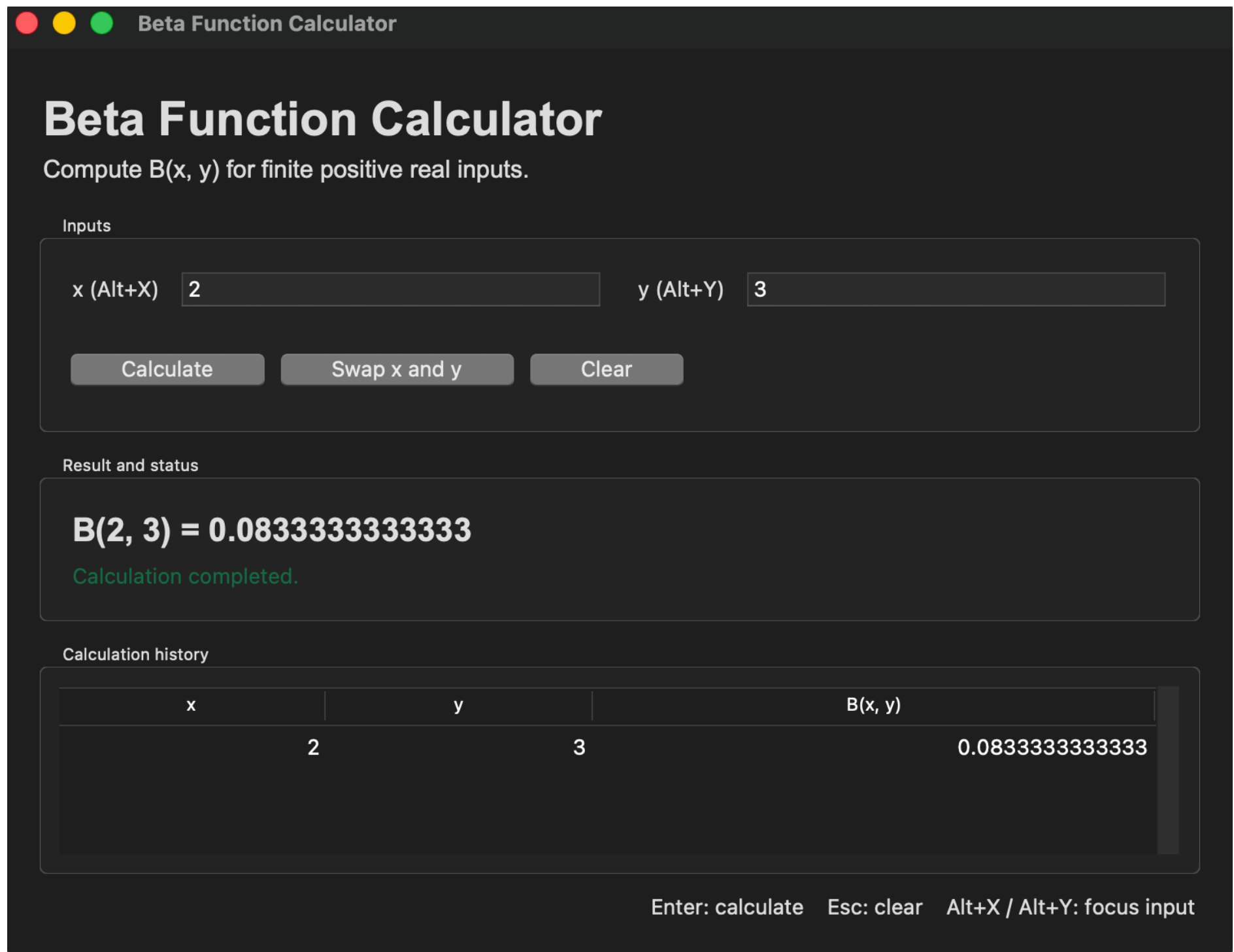


$$B(x,y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}, \quad x > 0, y > 0$$

Final Graphical Interface



The live interface demonstrates a known value, visible calculation status, and persistent history. Native Tkinter widgets preserve familiar focus and tab behavior.

Accessibility and Recovery

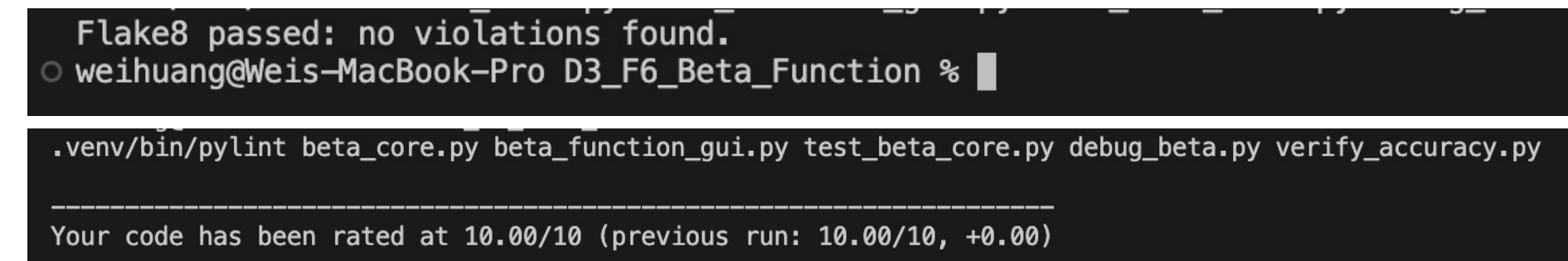
- Keyboard access
Enter, Esc, Alt+X, Alt+Y
- Recognition over recall
visible labels and actions
- Helpful recovery
field-specific text messages
- Status is not color-only
words identify success or error

The GUI aims to be accessible through readable labels, visible focus, redundant text feedback, keyboard operation, and recoverable validation. These choices align the interface with the user’s task model and established usability heuristics (Kamthan, 2026c; Nielsen Norman Group, 2026).

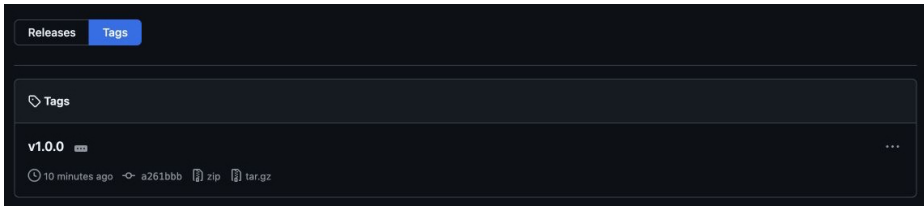
Exception policy: expected input and numeric-range errors receive specific messages; unexpected arithmetic failures are not hidden behind a generic message and remain available for diagnosis.

D3 conclusion: the release improves numerical stability, accessibility, and software quality through a stable log-beta implementation, explicit range behavior, and repeatable verification.

Problem 7: Quality and Versioning

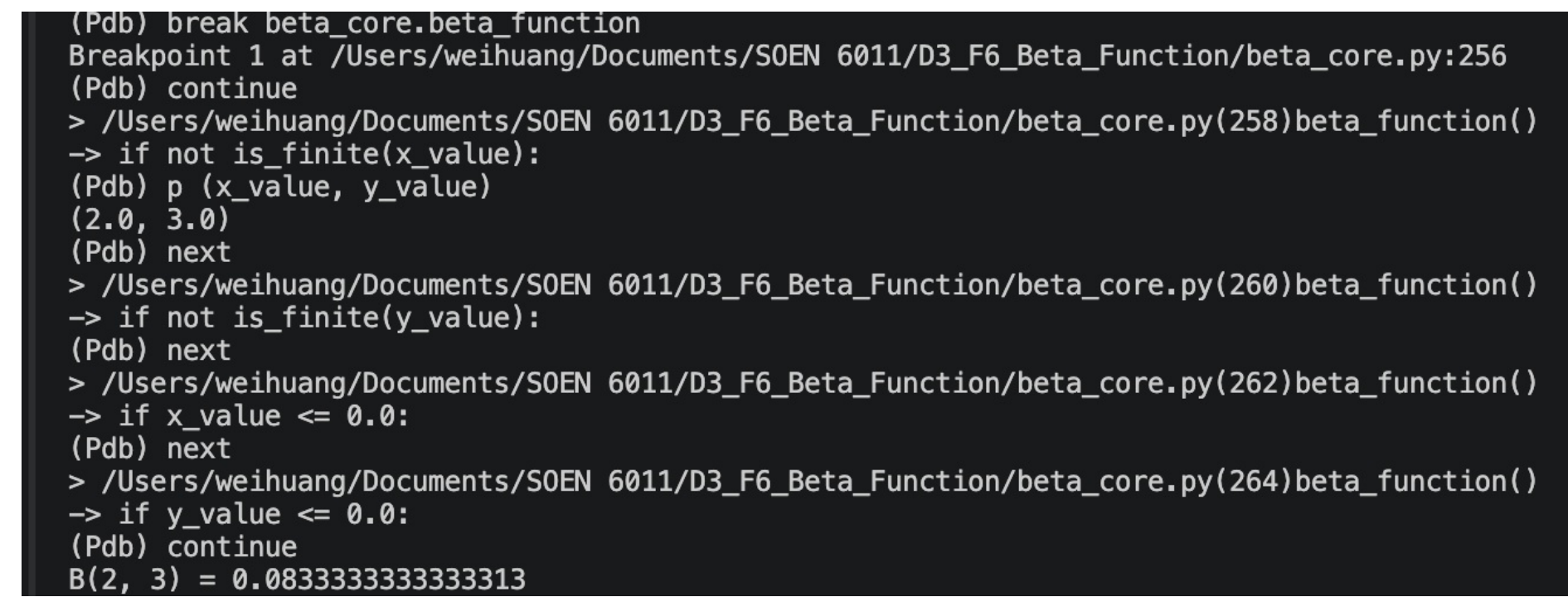


Correction made: Pylint reported too-many-public-methods after test expansion. One oversized test class was split into three focused classes, restoring **10.00/10** without a global suppression.



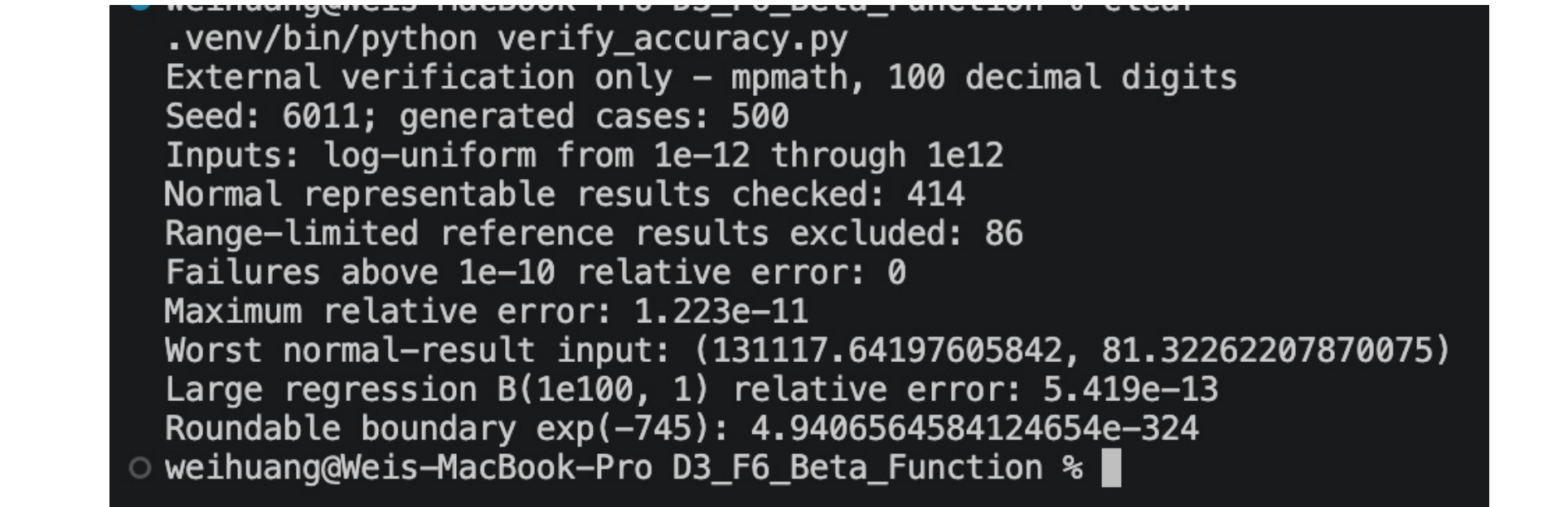
Why 1.0.0: first stable GUI and numerical contract, recorded by repository tag v1.0.0 (Semantic Versioning, 2026).

Problem 7: Debugger Evidence



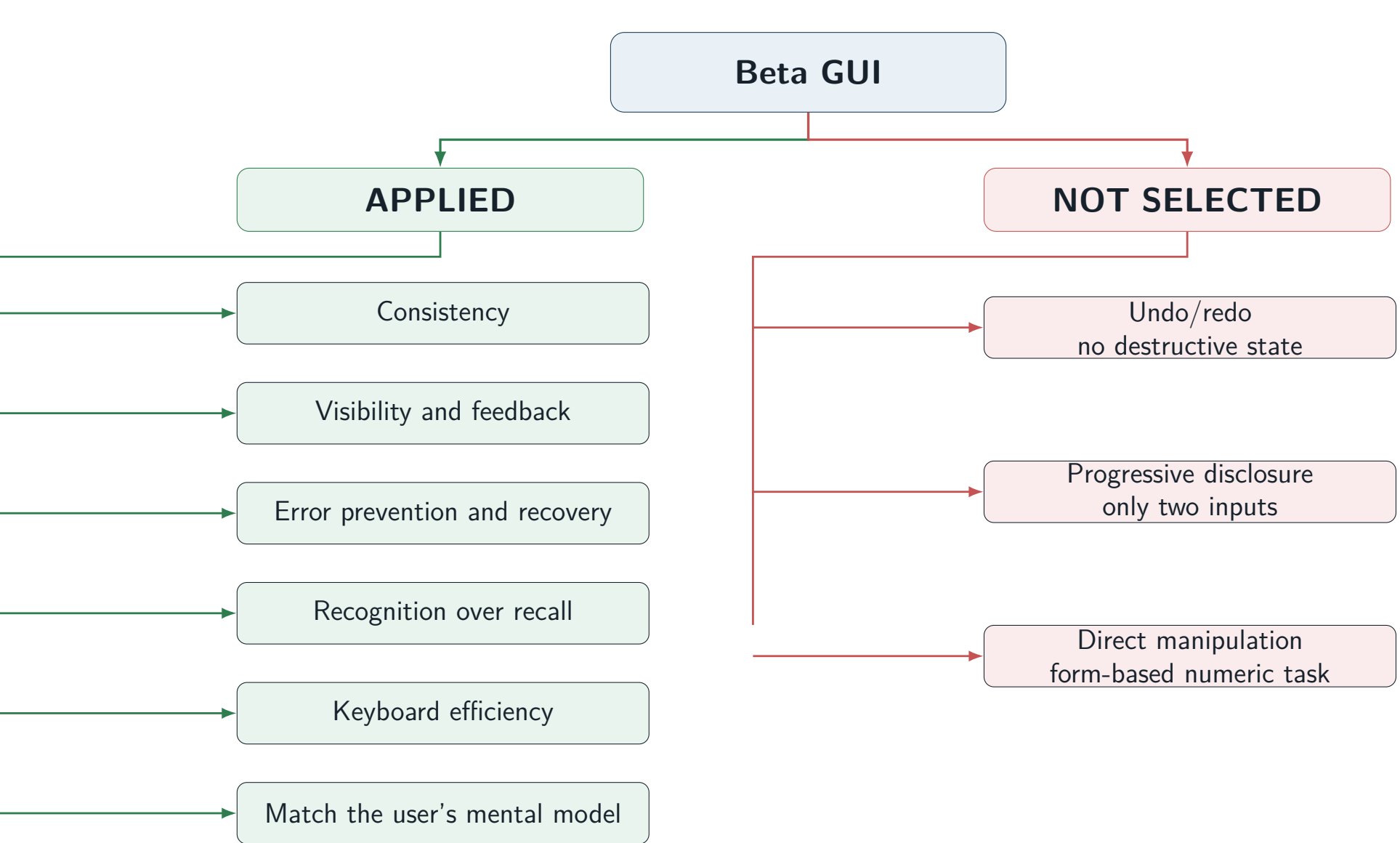
A reproducible pdb session stops inside beta_function, inspects (x, y) = (2, 3), and steps through finite and domain validation before continuing to the known result 1/12.

Verified Numerical Contract



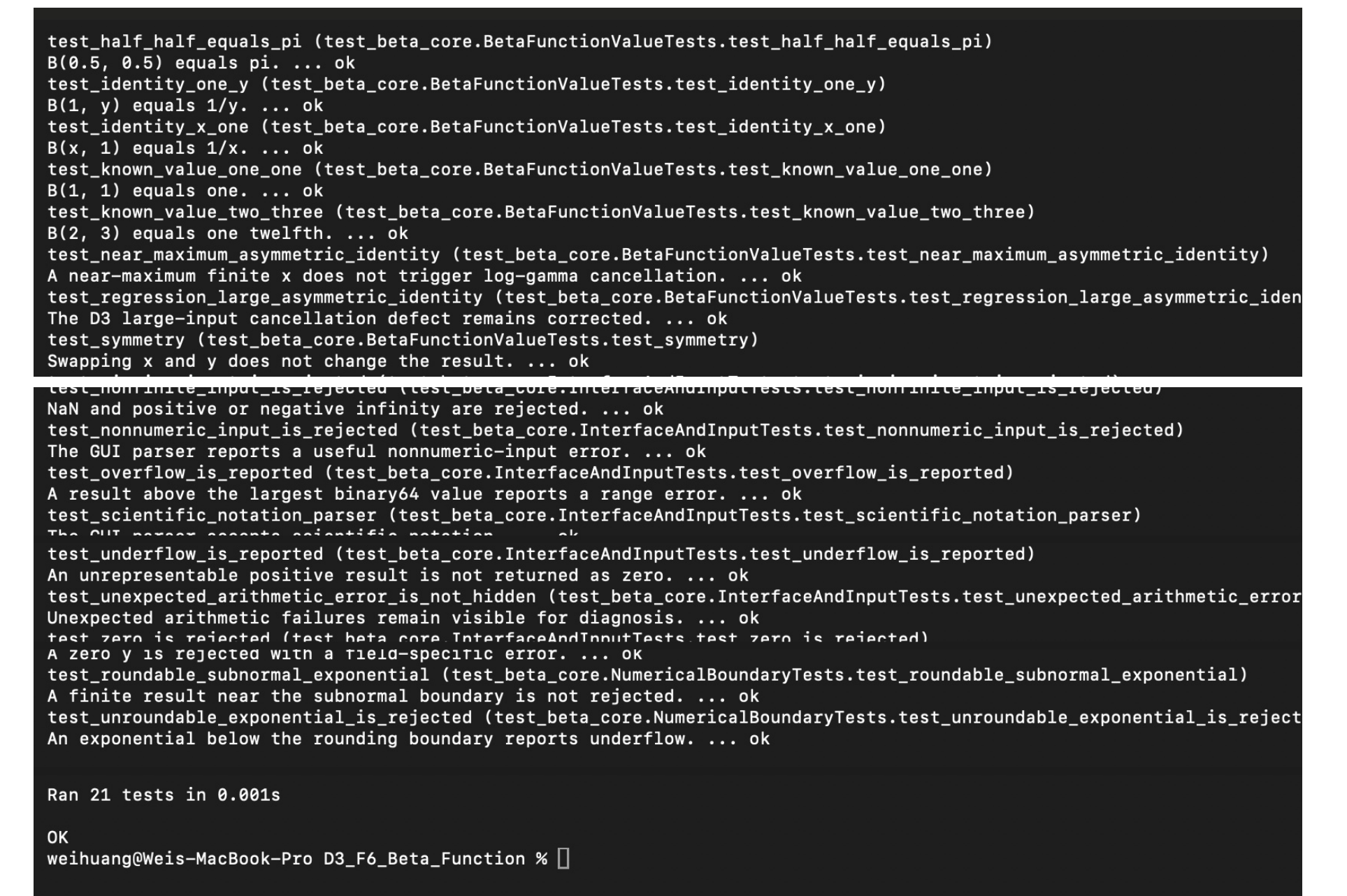
Contract: accepted inputs return a nonzero finite binary64 result when representable; otherwise the program reports overflow or underflow. Normal results in the deterministic oracle set meet 10⁻¹⁰ relative error; subnormals follow binary64 spacing (NIST DLMF, 2026; mpmath, 2023).

UIDP Decision Mind Map



The mind map records both selected and rejected principles instead of treating every principle as mandatory (Kamthan, 2026d; Nielsen Norman Group, 2026).

Problem 8: PyUnit Test Suite



21/21 tests pass. Coverage includes trusted values, $B(1,y) = 1/y$, $B(x,1) = 1/x$, symmetry, the D3 large-input regression, separate x/y domain errors, NaN/infinity, overflow, underflow, and both sides of the subnormal rounding boundary.

Generative AI and CASTROFF. Tool: OpenAI ChatGPT/Codex. **Constraints:** preserve from-scratch code and honest range claims; **Audience:** TA/instructor; **Structure:** evidence by Problems 7 and 8; **Tone:** concise and technical; **Role:** graduate student reviewer; **Output format:** code/test suggestions and poster statements; **Focus:** quality, accessibility, binary64 boundaries, and regression coverage; **Function:** identify missing evidence and unsupported claims. Example prompts requested a PEP 8 and range-behaviour review and focused PyUnit regressions. Suggestions were executed and revised; unverified claims were rejected. Full records are in GAI_USAGE.md.

References. Kamthan, P. (2026a), *SOEN 6011 Project Description*; Kamthan, P. (2026b), *Building, Versioning, and Releasing Software*; Kamthan, P. (2026c), *Introduction to Mental Models*; Kamthan, P. (2026d), *Brainstorming and Mind Mapping*; Nielsen Norman Group (2026), *Ten Usability Heuristics*; Semantic Versioning (2026), *Semantic Versioning 2.0.0*; NIST DLMF (2026), *Section 5.12: Beta Function*; mpmath (2023), *mpmath 1.3.0 Documentation*; Van Rossum, Warsaw, and Coghlan (2001), *PEP 8*; Python Software Foundation (2026), *unittest and pdb Documentation*.

Repository: github.com/HW925/SOEN-6011-D3-Beta-Function